

AYUSH KASHYAP

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Summary

Software Engineer with production experience building and debugging high-throughput services, ETL pipelines, and full-stack tools in Go, Python, TypeScript, JavaScript, and C++. Strong foundation in distributed systems, SQL, performance profiling, Git, Docker, and CI/CD, with hands-on C++ database engine contributions and distributed systems bug fixes. Experienced in developing automated, data-driven workflows and documenting production software.

Experience

Constructure AI | *Software Engineer*

Dec 2025 - Apr 2026

- Architected and debugged RESTful microservices in Go and Python, reducing p95 API latency from 1.8s to under 600ms through Redis caching, connection pooling, and async job queues; built high-throughput ETL pipelines that processed 500MB+ BIM/IFC datasets into PostgreSQL and pgvector with SQL-based validation, indexing, and query optimization.
- Designed and shipped a full-stack NL-to-SQL analytics engine using React, Next.js, TypeScript, and Node.js, backed by an agentic DAG orchestration system that coordinated four or more LLM agents across retrieval, SQL generation, validation, and summarization; reduced hallucinated outputs by 40% and enabled natural-language querying of complex construction databases.
- Implemented and validated a production RAG pipeline combining pgvector semantic search with structured SQL filtering, improving retrieval precision by 28%; performed software testing and debugging while deploying and monitoring services across staging and production under strict SLA requirements.

Open Source Contributions

Ray (ray-project/ray)

3+ merged PRs · PR #63952 · PR #64546

* Patched a **critical distributed systems bug** in Ray Core's C++ **OOM killer**: TimeBasedWorkerKillingPolicy killed the ServeController under memory pressure, triggering an infinite crash loop; plumbed an is_system_actor guard across **protobuf** → C++ **raylet** → **Cython** → **Python** with cross-boundary unit tests.

* Merged PR #64546 contributing to **KV-cache-aware LLM request routing** in Ray Serve - prefix-aware routing reduces redundant GPU recomputation across replicas, directly improving inference throughput for large-scale LLM deployments.

Milvus (milvus-io/milvus)

5+ merged PRs · PR #51051 · PR #50666 · PR #51636 (open)

* Implemented a new **Decimal scalar field type** end-to-end: protobuf wire format, Go schema parser, C++ **segcore** storage/indexing/query execution, and filter expression arithmetic - **1000+ line size/XXL PR**, all CI passing, under maintainer review.

* Extended the **filter expression engine** to support bitwise AND, OR, and XOR operators; modified full pipeline - protobuf, ANTLR parser, C++ **bitset engine**, and JSON/array/index execution paths (**500+ lines across 8 files**).

PROJECTS

GoChess Engine | [Demo](#)

- Built a chess engine in Go using data structures and algorithms including Minimax with Alpha-Beta Pruning, achieving 60% game-tree reduction and depth-5 search in under 200ms; cross-compiled it to WebAssembly with Web Workers for 2–3× browser throughput; integrated Zobrist Hashing, Transposition Tables with 100K+ cached positions, Quiescence Search, and PST-based board evaluation.

PyGit - Distributed Version Control Engine | [GitHub](#)

- Re-implemented core Git functionality with Blob, Tree, and Commit objects, content-addressable storage, SHA-1 persistence, O(1) object lookup, a staging area, and commit DAG traversal; built a CLI that replicates init, add, commit, and log with full object serialization and compression.

Technical Skills

- **Languages**: Go, TypeScript, JavaScript, Python, C++, Java, C#, Node.js
- **Frontend**: React, Next.js, React Native, WebAssembly (WASM), Web Workers
- **Backend & Distributed Systems**: REST APIs, Microservices, PostgreSQL, Redis, ETL Pipelines, Async Processing, Distributed Systems, High-Throughput Services, SQL Optimization, Databases
- **Agentic AI & ML**: LangChain, LangGraph, RAG, OpenAI SDK, MCP-style Agent Orchestration, pgvector, Milvus, Prompt Engineering, Multi-Agent Systems
- **DevOps & Infra**: Git, Docker, CI/CD, Ray, Performance Profiling, Observability, Software Testing And Debugging
- **Computer Science Fundamentals**: Data Structures And Algorithms, Operating Systems

EDUCATION

Punjab Engineering College (PEC), Chandigarh | *B.Tech, Mechanical Engineering* (GPA: 8.23 / 10) Chandigarh 2023 - 2027